Tissue Alterations in Chronic Wounds

It should also be recognized that tissues around chronic wounds are not normal and are altered by the primary pathogenic mechanisms or by the inability to heal readily. Clinically, the best example of this is the intense fibrosis surrounding venous ulcers, termed **lipodermatosclerosis**. Once established, lipodermatosclerosis can lead to ulcer formation and becomes a site of frequent ulcer recurrence. Venous ulcers surrounded by lipodermatosclerosis are notably more difficult to heal.

This may be due to alterations in the cellular composition of non-healing wounds. Most evidence to date—largely because of the ease of culturing—focuses on wound fibroblasts. Ulcer fibroblasts are known to be **senescent** and unresponsive to certain cytokines and growth factors. For instance, venous ulcer fibroblasts show unresponsiveness to **TGF-β1** and **PDGF**. The impaired response to TGF-β1 is associated with decreased expression of type II TGF-β receptors, leading to reduced phosphorylation of key signaling proteins such as **Smad2**, **Smad3**, and **MAPK**.

Similarly, the synthetic activity of cells in diabetic ulcers may be dysregulated, resulting in chronic wounds being "stuck" in a particular phase of the repair process. Several of these cellular abnormalities have been closely linked to impaired healing capacity.

Basic Standards of Care

Arterial Ulcers

- **Vascular reconstruction** is essential for healing arterial ulcers. Without revascularization, interventions are temporary and often culminate in amputation, especially in inoperable patients.
- Necrotic, stable arterial ulcers with an overlying eschar should not be disturbed unless a vascular reconstructive plan is in place.

Venous Ulcers

- **Leg elevation** and **compression therapy** are the cornerstones of treatment (see Chap. 175).
- **Contact dermatitis** is common in patients with venous disease; therefore, minimizing the use of topical agents is often necessary.
- **Pentoxifylline** has been evaluated in randomized trials for accelerating venous ulcer healing. Results are mixed, though higher doses (800 mg three times daily) may be more effective than the standard 400 mg three times daily. Its role as standard therapy remains uncertain.
- **Stanozolol**, an anabolic steroid, was effective in reducing induration in acute, painful lipodermatosclerosis when compression is intolerable, but it is no longer available.
- To prevent recurrence, **graduated elastic compression stockings** delivering ~40 mm Hg pressure at the ankle are currently the only proven medical intervention. These should be worn long-term to reduce ulcer recurrence and manage other manifestations of chronic venous disease.

Diabetic Foot Ulcers

- The primary pathophysiological triad—**neuropathy, ischemia, and trauma**—drives the development and recurrence of foot ulcers in diabetes (see Chap. 152).
- **Glycemic control** and management of systemic complications (e.g., renal failure) are critical components of therapy.
- **Off-loading** is the cornerstone of local wound care for neuropathic ulcers.
- Patients with ulcers due to vascular insufficiency or mixed etiology (neuropathic and ischemic) require **prompt referral** for vascular assessment and surgical consultation.